Trainings

General Information

Daily or Refueling - Maintenance Check (1)

- Fuel-Water Separator - drain
- Lubricating Oil Level - check
- Coolant Level - check
- Fan, Cooling - check
- Drive Belts - check
- Air Intake Piping - check
- Charge-Air Piping - check
- Air Tanks and Reservoirs - drain
- Crankcase Breather Tube - check

Every 250 Hours, or 6 Months[1, 4]

- Lubricating Oil and Filters - change
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - check
- Fuel Filter (Spin-On Type) - change

Every 1,500 Hours, or 1 Year[2,3]

- Coolant Filter - change
- Coolant Filter Head - Inspect for reuse
- Cooling Fan Belt Tensioner - check
- Air Leaks, Air Intake and Exhaust Systems - check
- Air Cleaner Restriction - check
- Engine Wiring Harness - check

Every 6,000 Hours, or 2 Years[3]

- Crankcase Breather Tube - check
- Radiator Hose - check
- Cold Weather Starting Aids - check
- Engine Steam Cleaning - clean
- Engine Mounting Bolts - check
- Vibration Damper, Viscous - inspect for reuse
- Overhead Set - adjust
- Crankcase Breather (Internal) - replace

Every 10,000 Hours, or 5 Years[3]

- Fan Hub, Belt Driven - replace
- Air Compressor Discharge Lines - check

1. The lubricating oil drain intervals can be adjusted based on fuel consumption and engine duty cycle. See the Oil Drain Interval section of this procedure for more details.
2. Follow the manufacturer's recommended maintenance procedures for the starter, alternator, generator, batteries, electrical components, charge-air cooler, air compressor, air conditioner compressor, and fan clutch
3. Perform maintenance at whichever interval occurs first. At each scheduled maintenance interval, perform all previous maintenance checks that are due for scheduled maintenance.
4. Test the SCA concentration level every 6 months unless the concentration is over three units; then check at every oil drain interval until the concentration is below three units. For coolant condemnation limits, Refer to Procedure 018-004 in Section V. (/qs3/pubsys2/xml/en/procedures/00/00-018-004.html)

Oil Drain Intervals

All Others Without EGR

Select an oil drain interval based on oil classification for Recreational Vehicle, Refuse, Mixer, Dump, Delivery, Logging, Fire Truck, or Crane vehicle applications.

Oil Classification	**Kilometers**	**Miles**	**Hours**	**Months**
CES 20071	11,500	7000	300	6
CES 20076 and CES 20078, CES 20081[1]	14,500	9000	400	6

If the application is **not** one of the above, select the oil drain interval Extreme Severe Duty, Severe Duty, Normal Duty, or Light Duty based on how you use the engine. See Oil Drain Intervals by severity km [mi] located in this section.

- Follow oil drain interval Extreme Severe Duty if the vehicle operates under either of the conditions listed in Extreme Severe Duty.
- Follow oil drain interval Severe Duty if the vehicle operates under either of the conditions listed in interval Severe Duty.
- Follow oil drain interval Normal Duty if the vehicle operates under either of the conditions listed in interval Normal Duty and does **not** meet any of the conditions listed in interval Severe Duty.
- Follow oil drain interval Light Duty if the vehicle operates under both of the conditions listed in interval Light Duty and does **not** meet any of the conditions listed in interval Severe Duty or interval Normal Duty.

1. When used with ultra-low sulfur diesel fuel (15 ppm sulfur), if the sulfur content of the fuel is greater than 15 ppm, the oil change intervals **must** be reduced by 20 percent.

Oil Classification	Extreme Severe Duty	Severe Duty	Normal Duty[3]	Light Duty[3]
	<1.27 km/liter [3 mpg]	**< 2.3 km/liter [5.5 mpg]**	**2.3 to 2.8 km/liter [5.5 to 6.5 mpg]**	**> 2.8 km/liter [6.5 mpg]**
CES 20071	7,000 km [4,500 mi]	24,000 km [15,000 mi]	56,500 km [35,000 mi]	72,500 km [45,000 mi]
CES 20076, CES 20078, CES 20081[1]	10,000 km [6,000 mi]	32,000 km [20,000 mi]	64,500 km [40,000 mi]	80,500 km [50,000 mi]

Note : Extending the oil and filter change interval beyond the recommendation will decrease engine life due to factors such as corrosion, deposits, and wear.

Note : If the sulfur content of the fuel is greater than 0.50 percent, the oil change intervals must be reduced by an additional 20 percent.

1. When used with ultra-low sulfur diesel fuel (15 ppm sulfur). If the sulfur content of the fuel is greater than 15 ppm, the oil change intervals **must** be reduced by 20 percent.
2. For Normal and Light duty cycles **only** (outlined in the above table): If Valvoline Premium Blue™ or Valvoline Premium Blue Extreme™ is being used; the oil drain intervals listed in the above chart can be increased by 8,050 km [5,000 miles]. All other oil brands **must** follow the oil drain intervals listed above.

The oil drain intervals for industrial engines are based on the duty cycle (as reflected by fuel consumption) and lubricating oil quality. The table below specifies the maximum oil drain interval for the listed lubricating oil classifications based on the three different duty cycles: Heavy, Medium, and Light.

- Follow oil drain interval Heavy Duty if the equipment uses more than 57 liter [15 gal] of fuel per hour.
- Follow oil drain interval Medium Duty if the equipment uses between 42 to 57 liter [11 to 15 gal] of fuel per hour.
- Follow oil drain interval Light Duty if the equipment uses less than 42 liter [11 gal] of fuel per hour.

Note : Extending the oil and filter change interval beyond the recommendation will decrease engine life due to factors such as corrosion, deposits, and wear.

12 Gallon Oil Drain Interval - Duty Cycle (Fuel Consumption)			
Oil Classification	**Heavy >57 liters/hour [15 gallons/hour]**	**Medium 42 to 57 liters/hour [11 to 15 gallons/hour]**	**Light < 42 liters/hour [11 gallons/hour]**
API CD-4, CE-4, CF-4[1, 3]	125	250	375
API CG-4[3]	250	375	500
API CH-4[3]	400	525	650
CES 20076, 20078, 20081/CH-4[2, 3]	500	625	750

14 Gallon Oil Drain Interval - Duty Cycle (Fuel Consumption)			
Oil Classification	**Heavy >57 liters/hour [15 gallons/hour]**	**Medium 42 to 57 liters/hour [11 to 15 gallons/hour]**	**Light < 42 liters/hour [11 gallons/hour]**
CES 20078/CI-4/ACEA E7	580	730	875
CES 20081/CJ-4/ACEA E9/JAMA DH-2	580	730	875

22 Gallon Oil Drain Interval - Duty Cycle (Fuel Consumption)			
Oil Classification	**Heavy >57 liters/hour [15 gallons/hour]**	**Medium 42 to 57 liters/hour [11 to 15 gallons/hour]**	**Light < 42 liters/hour [11 gallons/hour]**
CES 20078/CI-4/ACEA E7	910	1140	1375
CES 20081/CJ-4/ACEA E9/JAMA DH-2	910	1140	1375

1. The oil classifications CD, CE, and CF have been obsoleted by API and **must not** be used, as their specifications are no longer controlled.
2. Valvoline Premium Blue™ and Premium Blue™ 2000 meet CES 20076 standards.
3. Use the followng procedure for the lubricating oil filter specification table. Refer to Procedure 018-003 in Section V. (/qs3/pubsys2/xml/en/procedures/101/101-018-003.html)

The table below lists typical duty cycles by application.

Note : The actual duty cycle can vary from the chart below. In those cases, it is necessary to change the lubricating oil as a function of average fuel consumption. Therefore, select a column based on the representative fuel consumption range.

Typical Duty Cycles by Applications		
Heavy	**Medium**	**Light**
Air Compressor	Articulated Dump Truck	Crane
Combine	Irrigation Equipment	Rear Dump Truck
Dozer	Scraper	
Dragline	Skidder	
Excavator		
Farm Tractors		
Forage Harvester		
Rock Drill		
Tub Grinder		

For Generator Drive engines, this service interval is based on load factor (as reflected by fuel usage), lubricating oil quality, lubricating system capacity, and operating speed 1,500 rpm (50 Hz) or 1,800 rpm (60 Hz). Premium grade oils (API CG-4, CH-4, and CES 20076) are recommended for the QSX15 engine. The oil grades CD, CE, and CF have been obsoleted by API and **must not** be used, as their specifications are no longer controlled. There are two recommended methods for determining the proper oil change interval:

- Fixed hour method; based on fixed hours of operation or months of service, whichever occurs first.
- Chart method; based on known fuel consumption rates.

If the chart method is **not** used or, for all stand-by power applications, the oil **must** be changed at a regular interval or 12 months, whichever occurs first:

Application	Operating Speed	Sump Size	Change Interval
Standby power	1,500 rpm (50 Hz)	45 liters [12 gal]	125 Hours or 12 Months
All other applications		45 liters [12 gal]	250 Hours or 12 Months
Standby power	1,500 rpm (50 Hz)	53 liters [14 gal]	145 Hours or 12 Months
All other applications		53 liters [14 gal]	290 Hours or 12 Months

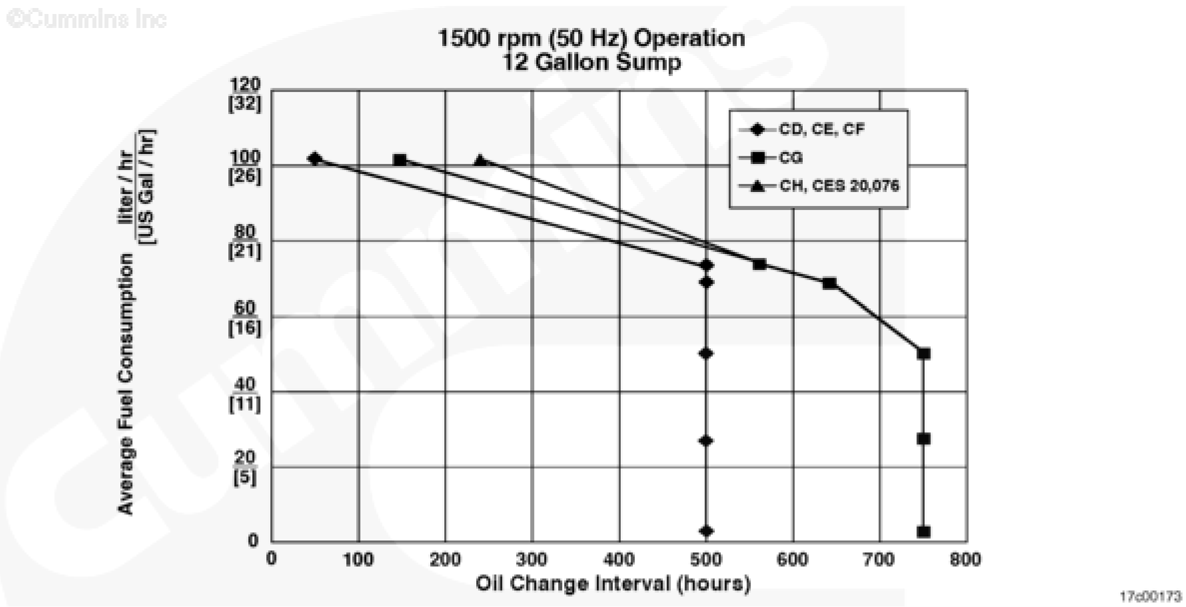
Application	Operating Speed	Sump Size	Change Interval
Standby power	1,800 rpm (60 Hz)	83 liters [22 gal]	250 Hours or 12 Months
All other applications		83 liters [22 gal]	500 Hours or 12 Months

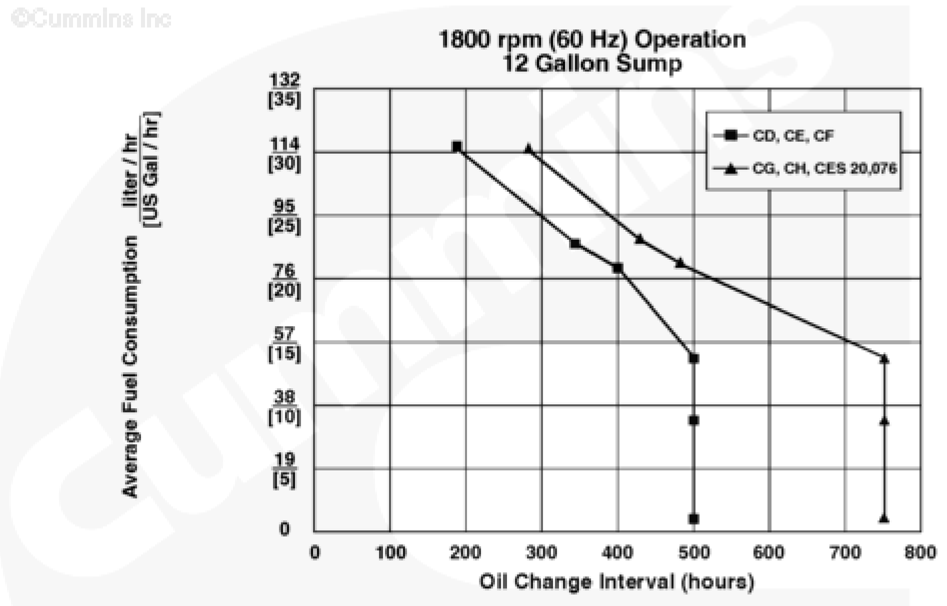
The chart method is recommended to provide the lowest total cost of operation while still protecting the engine. Due to differing availability outside North America, lower grade oil (CD, CE, and CF) are also depicted, however their classifications have been obsoleted by API, and oil change intervals are greatly reduced.

The charts **must** be used as guidelines because actual oil drain intervals will also depend on operation and maintenance practices. It is suggested that oil analysis **must** be used periodically for prime power applications (every 100 hours) to make sure the proper oil change interval is being applied.

To use the charts, locate the chart for the appropriate sump size and operating speed. Find the fuel consumption rate in liters per hour or U.S. gallons per hour on the left vertical axis. Draw a horizontal line from left to right across the chart, parallel with the bottom of the chart, until it intersects the curve.

From the intersection point on the curve, draw a line perpendicular to the bottom of the chart. The number the line intersects across the bottom of the chart represents the recommend oil change interval in hours.





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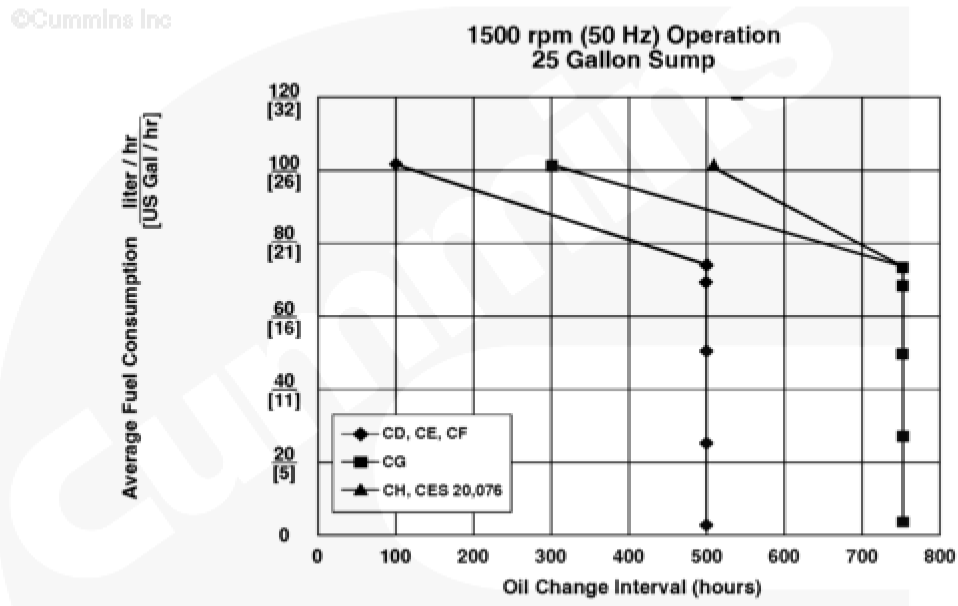

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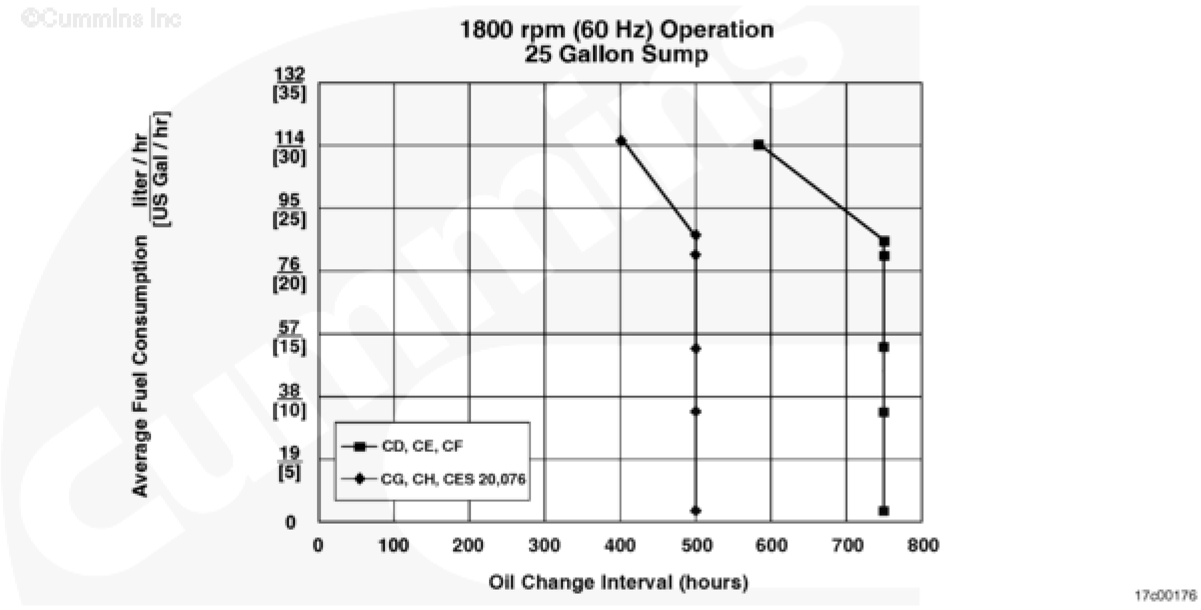

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